

MSCI World Factor Strategy

Portfolio backtest · Research report · 7 October 2026 USD results · 3 October 2014-28 August 2026

Abstract

This study evaluates a developed-market equity portfolio with Core, Momentum, Quality and Value weights of 60/15/10/15. Absolute ± 5 -percentage-point sleeve bands are combined with separate leverage management at a target of 1.25x. Over the common historical sample, CAGR is **11.87% without borrowing** and **13.24% at the borrowed target**, after modelled investment trading and financing costs. The borrowed portfolio has a 0.4190 pp CAGR advantage over the MSCI World/Core comparator at the same target exposure, with slightly lower observed volatility and drawdown. These favourable full-period comparisons coexist with changing period rankings and alpha intervals that include zero. Historical specification testing and unverified real lending conditions limit prospective conclusions. [Return and risk metrics: historical metrics](#).

1. Research Question

Can a portfolio anchored in the MSCI World, with momentum, quality and value sleeves, improve historical growth and risk outcomes relative to a broad-market investment? Does borrowing add return after investment costs, and what additional risks accompany it?

The main strategy uses 1.25x target leverage. Unlevered results provide a control for the contribution and risk of borrowing. A favourable joint historical comparison requires higher CAGR, no higher volatility and no deeper maximum drawdown. This criterion is distinct from statistical alpha or a reliable forecast of future outperformance.

2. Portfolio Design and Benchmark

2.1 Allocation and trading rules

Table 1. Target allocation · [Target weights: backtest configuration](#)

Sleeve	Target	Economic role
Core	60.00%	Broad developed-market anchor
Momentum	15.00%	Relative-performance selection
Quality	10.00%	Financial-quality selection
Value	15.00%	Valuation selection

The Core provides broad developed-market equity exposure; the factor sleeves apply different selection and weighting characteristics within the MSCI World framework. They remain equity investments and can decline together. The weights express the proposed investment design rather than a fitted estimate of optimal future weights.

Weights are measured against gross invested assets. Each sleeve may drift by five percentage points above or below its target: Core 55-65%, Momentum 10-20%, Quality 5-15% and Value 10-20%. Equality at a boundary does not trigger trading. A strict breach signals a full sleeve reset at the next common NAV observation. A sleeve-only reset preserves debt; a leverage-only adjustment scales the existing mixture proportionally. A simultaneous signal resets both.

Leverage is managed separately at 1.25x with a ± 0.10 band. This is an adjustment rule, not a hard cap on realised leverage. An account begins with USD 7.00 million of committed equity, corresponding to USD 8.75 million of gross exposure and USD 1.75 million of debt before opening charges. [Portfolio and cost settings: backtest configuration](#).

Five-percentage-point bands were the author's initial band-width proposal. Tighter alternatives were explored before returning to that width for simplicity and greater permitted factor drift. This chronology does not make the final specification independent of the historical evidence. The separated leverage mechanism must also be distinguished from coupled resets in some exploratory controls. A capped-relative rule, using the smaller of five percentage points and twenty percent of each sleeve target, is retained as a sensitivity in Appendix A.

2.2 Why MSCI World is the benchmark

The MSCI World is the primary market benchmark because it supplies the common developed-market framework for the Core and factor sleeves. A broad-market investment is the natural alternative against which to evaluate the combined allocation and management rules. The 60% Core allocation reinforces this connection; the common framework is the principal justification.

The backtest implements this benchmark through the **iShares Core MSCI World UCITS ETF**, an investable proxy whose NAV includes embedded fund expenses. It is not the bare index return. Results compare both unlevered investments and investments at the same 1.25x target. Equal target exposure does not establish equal realised leverage, beta, costs or investment universes.

Dimensional is a secondary systematic-investment alternative. Amundi is a short-period real leveraged-product comparison. Neither replaces the MSCI World/Core benchmark.

3. Data and Method

3.1 Inputs and common calendar

Table 2. Main analytical inputs · [NAV and funding inputs: backtest_sources.json](#)

Source	Period	Use
BlackRock/iShares accumulating ETF NAV exports	Common sample: 3 October 2014-28 August 2026	Core, Momentum, Quality and Value investment returns
Author-supplied Bloomberg Dimensional NAV hardcopy	Same common sample	Secondary systematic-fund comparison
New York Fed indicative reference series and official SOFR	Relevant financing dates	Reference rate for borrowed exposure
Saved Amundi NAV workbook	Separate overlap: 30 September 2025-31 August 2026	Real daily 2x product comparison

The main common calendar contains **2929 levels and 2928 adjacent return intervals**. Investment levels are intersected before calculating returns; missing investment NAVs are not filled. The short Amundi experiment uses its own overlap and extends to August month-end. Reference rates are carried across nonpublication calendar days. The early indicative series is distinct from later official SOFR; both are financing proxies rather than the actual account's borrowing rate. [Common NAV calendar: backtest_workflow_prepare_inputs](#).

Accumulating NAVs include embedded fund expenses and reinvestment treatment. Deducting TER again would double-count that layer. NAV is not an executable bid or ask. Investor trading costs are therefore specified separately. Published rounded issuer returns provide a secondary check for 40 calendar-year observations; the

largest central-value difference is 0.0710 pp and remains compatible with the retained rounding assumptions. This is not a second complete daily-price feed. [Issuer return check: backtest_issuer_returns.run_issuer_returns](#).

The Dimensional comparison uses the Global Core Equity Fund, USD Accumulation Shares (ISIN IE00B2PC0153), with NAV observations from the supplied Bloomberg workbook. Its investment mandate differs from MSCI World and allows some emerging-market investments. That permission does not establish the fund's actual emerging-market allocation throughout the sample. Fund identities and sources are listed after the conclusion.

3.2 Accounting and costs

AI-assisted methodology and writing: OpenAI Codex (OpenAI, n.d.) assisted development of the analytical code and all narrative sections, tables and figures. The final AI Assistance and Responsibility section identifies the affected material, tools, author decisions, checks and retained prompt evidence.

Gross invested assets minus debt equals investor equity. At an observation, positions are marked and financing is accrued; prior signals are executed before new signals are evaluated. Trading charges reduce equity. No subscriptions or withdrawals enter the investment backtest. [Trading, debt and fees: historical.simulate](#).

Table 3. Investment cost and measurement assumptions · [Trading and borrowing costs: backtest configuration](#)

Baseline parameter	Selected value
Credit markup	3.00%
Reference day count	ACT/360
Platform fee per traded leg	USD 0.85
Investor-side duty	0.15%
Ordinary one-way spread	5 bp
USD per CHF	1.2367747440

Funding accrues from the previous observation inclusive to the current observation exclusive, on ACT/360, and is capitalised at NAV dates. Current-cost scenarios apply the stated tariff and commission conversion throughout history rather than reconstructing historical invoices. Fund administration charges, custody sensitivities, investor taxes and manager business budgets are outside the baseline investment account. [Trading, debt and fees: historical.simulate](#).

The fixed commission conversion is **1 CHF = USD 1.2368**, derived from the recorded ECB reference rates for **31 August 2026**: 1.1596 USD per EUR divided by 0.9376 CHF per EUR. Only the CHF-denominated broker commission is converted; investment returns and debt already use USD. Full precision is retained in the calculation. This endpoint reference rate is a modelling assumption, not a historical series of broker execution rates. [Portfolio and cost settings: backtest configuration](#).

CAGR uses committed opening capital and elapsed calendar years. Opening trading charges are included once and folded into the first investor-return interval. Drawdown uses a high-water mark no lower than committed opening capital. Volatility uses 252-observation annualisation. Jensen alpha is an annualised excess-return regression intercept against unlevered Core, not the CAGR difference; exploratory intervals use five-lag HAC errors. [Return and risk metrics: historical.metrics](#).

Separate implementations check account identities and selected daily, event and outcome results under shared inputs and conventions. These checks address implementation errors; they do not establish market-price executability, commercial feasibility or expert peer review.

4. Historical Results

4.1 Central MSCI World comparison

Table 4. Unlevered control over the main sample · [CAGR, volatility and drawdown: historical.metrics](#)

Investment / rule	CAGR	Volatility	Drawdown	Sharpe	Ending equity, USD m
Absolute / separate leverage	11.87%	14.96%	-33.56%	0.702	26.59
MSCI World Core	11.52%	15.18%	-34.07%	0.672	25.62

Table 5. Main portfolio and Core at a 1.25x target · [CAGR, volatility and drawdown: historical.metrics](#)

Investment / rule	CAGR	Volatility	Drawdown	Sharpe	Ending equity, USD m
Absolute / separate leverage	13.24%	18.59%	-39.70%	0.665	30.76
MSCI World Core	12.82%	18.94%	-40.36%	0.636	29.43

The main strategy meets the stated joint growth/risk criterion against Core over the full sample. Its CAGR advantage is 0.3503 pp without borrowing and 0.4190 pp at matched target leverage. These are historical account comparisons, not a causal estimate of the return from each factor. [Return and risk metrics: historical.metrics](#).

Borrowing increases portfolio CAGR from 11.87% to 13.24%, but also raises volatility from 14.96% to 18.59% and deepens maximum drawdown from -33.56% to -39.70%. The additional growth comes with additional total risk. [Return and risk metrics: historical.metrics](#).

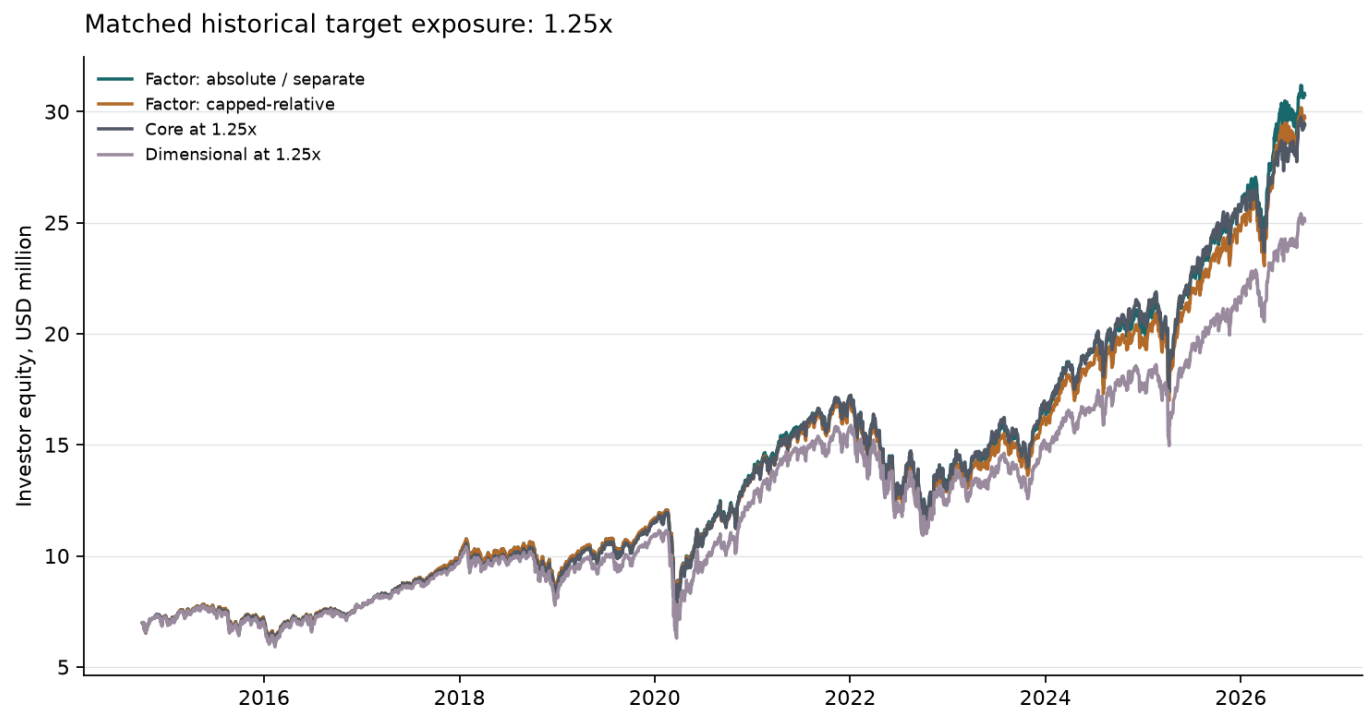


Figure 1. Investor equity at matched 1.25x target exposure, after modelled trading and financing. The capped-relative sensitivity is shown alongside the main strategy. There are no subscriptions or manager-fee receipts in these paths. [Equity and drawdown plots: backtest_report.plot_history](#)

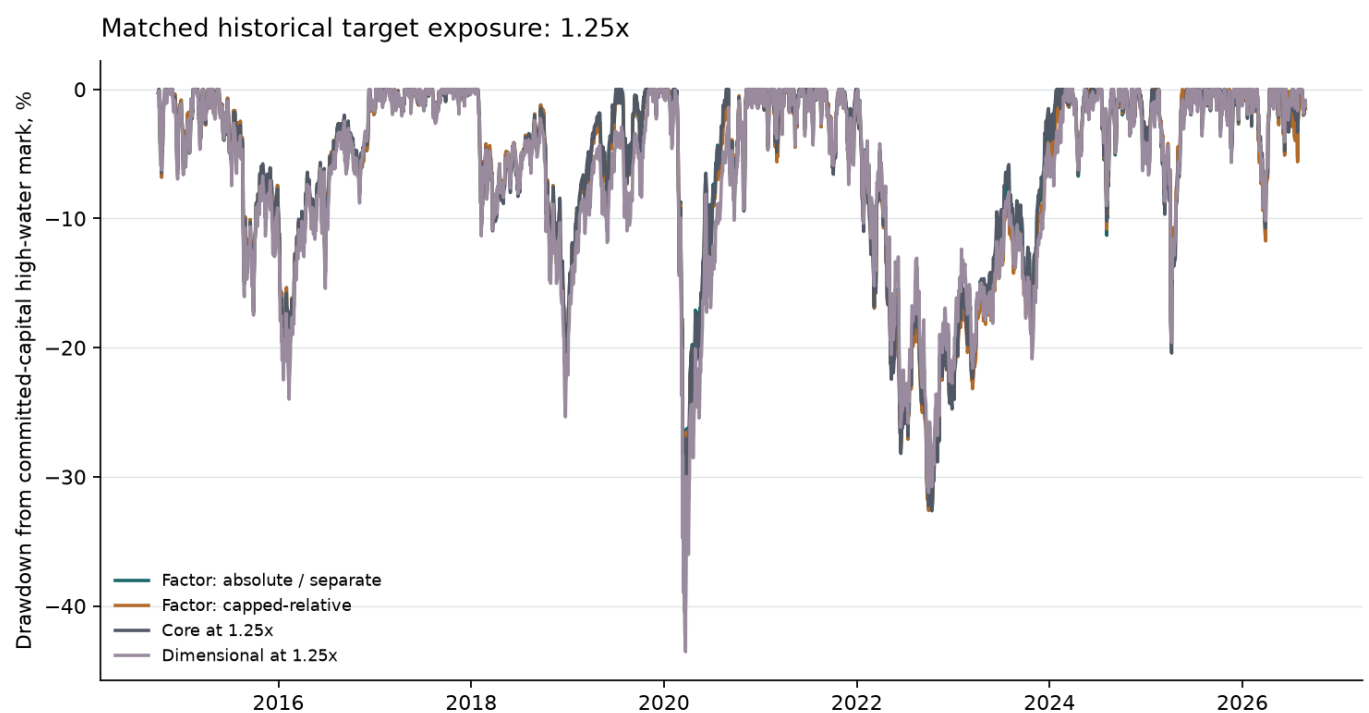


Figure 2. Drawdown from the committed-capital high-water mark on the common observation calendar. The paths include opening charges but do not capture intraday lender exposure. [Equity and drawdown plots: backtest_report.plot_history](#)

4.2 Brief comparison with Dimensional

Table 6. Main strategy and Dimensional at the 1.25x target · [CAGR, volatility and drawdown: historical metrics](#)

Investment / rule	CAGR	Volatility	Drawdown	Sharpe	Ending equity, USD m
Absolute / separate leverage	13.24%	18.59%	-39.70%	0.665	30.76
Dimensional	11.32%	20.92%	-43.55%	0.529	25.08

The main portfolio's CAGR advantage over Dimensional is 1.9262 pp, with lower observed volatility and a shallower maximum drawdown. Dimensional is an existing systematic fund alternative with different universe and implementation characteristics. Equal target borrowing does not remove those differences. The comparator control excluding external transaction charges also remains below the primary over the full sample; embedded product costs remain in that control. This comparison therefore does not isolate a pure factor premium. [Return and risk metrics: historical metrics](#).

4.3 Portfolio at 2x versus the Amundi daily 2x ETF

Table 7. Factor portfolio at 2x and Amundi daily 2x: 30 September 2025-31 August 2026 · [Product comparison: backtest_products.product_comparison](#)

Short-overlap investment	Cumulative return	Volatility	Drawdown
factor absolute decoupled 2.00x	36.38%	26.22%	-18.74%
amundi 2x	27.51%	23.79%	-17.50%

For this product comparison, the factor portfolio uses a **2x leverage target** with the same allocation and absolute sleeve bands. Its debt is adjusted through the separate leverage band. It is compared with the **Amundi daily 2x ETF** over their common short window. The table reports cumulative returns. The factor portfolio earns the higher return, accompanied by higher volatility and a deeper maximum drawdown. The main strategy and its MSCI World/Core benchmark comparison continue to use **1.25x**.

The factor portfolio includes modelled ongoing trading and borrowing costs. The Amundi return is taken directly from its USD NAV, which incorporates the product's expenses and leveraged financing economics. **The portfolio's borrowing charges are therefore not applied to Amundi.** The Amundi series excludes additional investor purchase/exit brokerage, taxes and listing spreads. Both paths are normalised to the first common post-entry observation; this removes the modelled portfolio's opening expenses.

Amundi resets exposure daily, while the factor portfolio manages leverage within a band. These mechanics produce different exposure paths despite the same nominal 2x label. The evidence is limited to the observed product overlap and does not establish long-period or future superiority. The 2x experiment remains an additional comparison; it leaves the main investment and capital-inflow specifications unchanged.

5. Robustness and Implementation Risk

The growth ranking is not stable across all dates. In independently restarted early and later periods, Core exceeds the main portfolio's CAGR in the early segment, while the main portfolio leads in the later segment. Appendix B retains this counter-evidence and the alpha estimates. Overlapping rolling windows and alternative entry dates do not provide independent probabilities of future success.

The main portfolio's exploratory alpha intervals include zero at both exposure levels. The leveraged estimate is not convincingly positive. Historical testing, changes in period rankings and a market-exposed set of equity sleeves prevent a claim of established future superiority. The later restart is retrospective rather than a pristine holdout.

Financing and custody costs can change signals, holdings and later trades. Consequently, small changes in charges need not produce monotone ending values. The financing sensitivity in Appendix B is descriptive; it does not identify a universally safe borrowing margin. Higher ending wealth in an isolated fee scenario is not evidence that paying higher fees is beneficial.

Hypothetical instantaneous losses at maintenance boundaries include both sale-funded cures and insolvency before a cure is possible. These are imposed account states with liquidation friction, not actual historical lender accounts. Positive equity and successful modelled sales do not establish that a lender permits the cure or executes at that NAV.

Actual collateral eligibility, account category, rate fixing, settlement, liquidation timing and complete charges remain unconfirmed. The public banking material supports scenario definitions rather than an individual credit offer. Sparse holdings snapshots cannot explain the entire historical return difference or supply a full historical sector panel.

6. Conclusion

Under the stated conventions, the 60/15/10/15 portfolio with absolute ± 5 -percentage-point bands and separate leverage management has a favourable full-sample growth/risk comparison with MSCI World/Core. The simpler common band consciously permits factor drift. Borrowing adds historical growth and materially increases total risk.

The evidence supports further investigation of a fixed specification rather than a forecast of superior returns. Prospective evaluation and verified lending and execution terms are separate requirements from reproducing the historical accounts. The capital-inflow companion report applies the portfolio to a shorter historical fund experiment with hypothetical fees, flows and business costs. Its separate flat, fixed-growth and imposed-loss controls are not projections of the full-period backtest CAGR.

Sources and Data

Fund identities: Core IE00B4L5Y983; Momentum IE00BP3QZ825; Quality IE00BP3QZ601; Value IE00BP3QZB59. Dimensional IE00B2PC0153; Amundi FR0014010HV4.

Market data and financing series

NAV observations and reference rates enter the calculations. Their sample dates are specified in the analysis. Provider links identify the funds and series; the supplied local files are the calculation inputs.

Source	Publisher	Use in this report
Core MSCI World NAV	BlackRock / iShares	Core accumulating USD NAV; investment returns
Momentum NAV	BlackRock / iShares	Momentum accumulating USD NAV; investment returns
Quality NAV	BlackRock / iShares	Quality accumulating USD NAV; investment returns
Value NAV	BlackRock / iShares	Value accumulating USD NAV; investment returns
Indicative reference rate	Federal Reserve Bank of New York	Early indicative financing reference series
Official SOFR	Federal Reserve Bank of New York	Later official financing reference series
Dimensional NAV hardcopy	Bloomberg, author-supplied hardcopy	USD fund NAV; secondary fund comparison
Amundi daily 2x NAV	Amundi	USD ETF NAV; short-window daily 2x product comparison

Cost, product and methodology references

These sources support the stated assumptions, product descriptions and checks. They are distinguished from time-series inputs.

Source	Publisher	Use in this report
Commission tariff	Swissquote	Current CHF commission schedule for the investment cost model
Investor-side stamp duty	Swiss Federal Tax Administration	Basis for the modelled 0.15% investor-side duty
Commission FX conversion	ECB / Official Journal	31 August 2026 reference cross-rate; CHF commission conversion only
Dimensional key information document	Dimensional Funds plc / FundAssist	USD share-class identity and permitted investment universe; 17 July 2025 KID
Reference-rate methodology	Federal Reserve Bank of New York	Interpretation of the early reference-rate proxy
Risk-measure definitions	CFA Institute	Background on risk-adjusted performance; calculation conventions stated in Method
Collateralised loan terms	Swissquote Bank Ltd Switzerland	Contractual context for hypothetical credit and liquidation risks
Lombard loan overview	Swissquote Bank Ltd Switzerland	Product context for borrowing; not an account-specific rate quote

Source	Publisher	Use in this report
Published Core annual returns; Published Momentum annual returns; Published Quality annual returns; Published Value annual returns	BlackRock / iShares	Rounded calendar-year returns; secondary validation of NAV calculations
Amundi key information document, 28 April 2026	Amundi	Daily-reset 2x mechanism and embedded product/financing costs; not a NAV download source

Original files and download records are retained with the reproducibility materials.

Appendix A. Band Sensitivity

Table A1. Absolute/separate and capped-relative policies · [Policy accounts: backtest_workflow.policy_comparison](#)

Rule	Exposure	CAGR	Volatility	Drawdown
Absolute / separate leverage	1.000	11.87%	14.96%	-33.56%
Capped-relative comparison	1.000	11.79%	14.98%	-33.72%
Absolute / separate leverage	1.250	13.24%	18.59%	-39.70%
Capped-relative comparison	1.250	12.92%	18.47%	-39.89%

The capped-relative rule has narrower factor bands, particularly for Quality. Broader bands do not guarantee lower overall risk or fewer trades on every path. Allocation grids and legacy coupled-absolute search maxima are retrospective sensitivity evidence and do not define the separated primary or an optimal future allocation.

Appendix B. Statistical and Period Checks

Table B1. Main portfolio alpha against unlevered Core · [Alpha and HAC intervals: historical.metrics](#)

Exposure	Beta	Alpha / year	HAC low	HAC high
1.000	0.980	0.49%	-0.35%	1.33%
1.250	1.219	-0.08%	-1.11%	0.95%

Intervals are exploratory nominal-252 annualised regression-intercept intervals with five-lag HAC errors. They are not adjusted for specification selection or multiple testing. The leveraged regression remains against unlevered Core, so its intercept must not be confused with the matched-target CAGR gap.

Table B2. Independently restarted retrospective periods at 1.25x · [Policy accounts:](#)[backtest_workflow.policy_comparison](#)

Retrospective period	Investment	Exposure	CAGR	Volatility	Drawdown
Early (retrospective)	Factor portfolio	1.250	13.05%	18.61%	-39.70%
Early (retrospective)	MSCI World Core	1.250	13.15%	18.91%	-40.36%
Later (retrospective)	Factor portfolio	1.250	12.61%	18.20%	-32.94%
Later (retrospective)	MSCI World Core	1.250	11.44%	18.49%	-33.09%

The early period is 3 October 2014-31 December 2021; the later period starts on the first common 2022 observation, 4 January, and ends on 28 August 2026. Each account restarts capital and opening charges. These splits preserve counter-evidence rather than provide an untouched holdout.

Table B3. Financing-markup sensitivity above the reference rate · [Borrowing scenarios:](#)[backtest_funding.funding_comparison](#)

Markup above reference, bp	Primary full CAGR	Primary later CAGR
0	13.64%	13.30%
300	13.24%	12.61%
500	12.77%	12.13%
1000	11.41%	11.11%

The later result is a restarted account, not a continuation of the full-history balance. Policy discontinuities prevent interpreting a finite scan as a continuous or globally complete break-even search.

Reproducibility Materials

Blue highlighted links name the source function or configuration and the quantity it calculates or specifies. Click a function link to open the corresponding code line on GitHub. Code links identify a fixed source snapshot; [the report-to-calculation map](#) provides the selected rows, fields and generated evidence paths. Original provider observations remain local.

The prepared code and separately supplied original data reconstruct the investment accounts. Full allocation, policy, rolling-entry, correlation, cost, artificial-regime, gap-cure, holdings and product-control results remain in the accompanying calculation materials. Their inclusion in the research archive does not make raw provider data public. The reader-selection record identifies the evidence behind each displayed table and claim.

AI Assistance and Responsibility

This portfolio-backtest and capital-inflow research package was developed throughout with extensive assistance from generative artificial intelligence. AI was used across the analytical and written workflow, including methodology development, programming, data processing, backtesting and scenario analysis, statistical evaluation, interpretation, report drafting and editing, tables, figures, and documentation. The original market data and external documents come from their respective providers and are referenced separately. The accompanying reproducibility materials document the calculations, assumptions, outputs, and checks. Responsibility for the submitted content remains with the human author(s).

Scope of assistance

OpenAI Codex was used for substantive code generation and revision, methodological discussion, analysis and report drafting. AI-generated or substantially revised wording appears throughout the abstract, main text, source commentary and appendices. Tables and figures were produced by AI-assisted code using the stated data and scenario assumptions. Provider data and external documents retain their original attribution.

List of aids (Hilfsmittelverzeichnis)

Tool	Purpose and affected material
OpenAI Codex (OpenAI, n.d.)	Method development, code generation/revision, data processing, backtests and scenarios, statistics, interpretation, all report prose, tables, figures and documentation
Python, NumPy, pandas and openpyxl	Numerical implementation, data alignment/accounting, scenario processing, original spreadsheet readers and report evidence
Matplotlib	Programmatic charts from calculated investment and fund paths
pytest	Automated calculation and report-contract checks
pypdf, pdfplumber, ReportLab, Pillow and Poppler	Source-PDF extraction/checks, report PDF generation and rendered-page verification

Author contribution and verification

Alex specified and approved key design choices, supplied project material, determined which comparisons to present and reviewed successive drafts. Codex helped translate these instructions into data readers, portfolio and fund simulations, analytical controls and reporting routines. It also contributed to the development of methods and interpretation of results.

Verification covered accounting identities, comparisons with retained reference outcomes, separate implementations, report evidence bindings and numerical, text and layout checks. These checks assess implementation and reporting under shared inputs and conventions; they do not constitute independent human expert review. The accompanying evidence record identifies the checks performed and their limits.

Documentation and archive

The accompanying AI-use disclosure record summarises the assistance, author contribution and verification limits. Selected exact prompt evidence and detailed internal working records are retained locally for applicable assessment/archive or review requirements; they are not distributed with the public reports.

A complete chronological prompt/output archive has not been established. Codex is identified at application level; no historical model version is asserted. Factual claims refer to the separately cited original evidence. The author remains responsible for the content and any assessment-specific documentation obligations.

AI application reference

OpenAI. (n.d.). *Codex* [Generative AI coding assistant]. Application reference.

This declaration concerns Alex's research contribution. It does not attribute unverified AI use to other group members or future video narration.